Data Science: SQL Assignment

SQL and DBeaver

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```
# Load required libraries
#| warning: FALSE
#| output: FALSE
invisible(lapply(c("DBI", "duckdb", "nycflights13"),
                 library, character.only=TRUE))
```

```
# Establish database connection
#| warning: FALSE
#| output: FALSE
connection <- DBI::dbConnect(drv = duckdb::duckdb(),
                             dbdir = "./data/flights.duckdb")
```

```
duckdb is storing downloaded extensions and secrets under ~/.duckdb:
i /Users/zakz/.duckdb
This persists across sessions and is shared with the DuckDB CLI and other clients.
i Run duckdb(shared_home = FALSE) to use a temporary directory instead.
i See ?duckdb_storage for details and alternatives.
```

Part I: Answer the following question by running SQL using RStudio.

Section 1.a: List the first 10 rows of the flights table.

```
# Get the first ten rows of the flights table
dbGetQuery(connection, "SELECT * FROM flights LIMIT 10")
```

```
   year month day dep_time sched_dep_time dep_delay arr_time sched_arr_time
1  2013     1   1      517            515         2      830            819
2  2013     1   1      533            529         4      850            830
3  2013     1   1      542            540         2      923            850
4  2013     1   1      544            545        -1     1004           1022
5  2013     1   1      554            600        -6      812            837
6  2013     1   1      554            558        -4      740            728
7  2013     1   1      555            600        -5      913            854
8  2013     1   1      557            600        -3      709            723
9  2013     1   1      557            600        -3      838            846
10 2013     1   1      558            600        -2      753            745
   arr_delay carrier flight tailnum origin dest air_time distance hour minute
1         11      UA   1545  N14228    EWR  IAH      227     1400    5     15
2         20      UA   1714  N24211    LGA  IAH      227     1416    5     29
3         33      AA   1141  N619AA    JFK  MIA      160     1089    5     40
4        -18      B6    725  N804JB    JFK  BQN      183     1576    5     45
5        -25      DL    461  N668DN    LGA  ATL      116      762    6      0
6         12      UA   1696  N39463    EWR  ORD      150      719    5     58
7         19      B6    507  N516JB    EWR  FLL      158     1065    6      0
8        -14      EV   5708  N829AS    LGA  IAD       53      229    6      0
9         -8      B6     79  N593JB    JFK  MCO      140      944    6      0
10         8      AA    301  N3ALAA    LGA  ORD      138      733    6      0
             time_hour
1  2013-01-01 10:00:00
2  2013-01-01 10:00:00
3  2013-01-01 10:00:00
4  2013-01-01 10:00:00
5  2013-01-01 11:00:00
6  2013-01-01 10:00:00
7  2013-01-01 11:00:00
8  2013-01-01 11:00:00
9  2013-01-01 11:00:00
10 2013-01-01 11:00:00
```

Section 1.b: Find the number of flights

i. in the dataset

```
# Get the total count of rows in the flights table
dbGetQuery(connection, "SELECT COUNT(*) FROM flights")
```

```
  count_star()
```

```
1       336776
```

ii. for each carrier

```
# Get the count of rows for each airline carrier
dbGetQuery(connection, "SELECT COUNT(*), carrier FROM flights GROUP BY carrier ORDER BY COUN
```

```
   count_star() carrier
1         58665      UA
2         54635      B6
3         54173      EV
4         48110      DL
5         32729      AA
6         26397      MQ
7         20536      US
8         18460      9E
9         12275      WN
10         5162      VX
11         3260      FL
12          714      AS
13          685      F9
14          601      YV
15          342      HA
16           32      OO
```

Section 1.c: Find the average departure delay (dep_delay)

```
# Get the mean of dep_delay from the flights table
dbGetQuery(connection, "SELECT AVG(dep_delay) FROM flights")
```

```
  avg(dep_delay)
1       12.63907
```

Section 1.d: Find the top 5 destinations (dest) With the most flights.

```
# Group by destinations, count the number of observations for each destination,
# order by the counts in descending order, and limit the output to the top five.
dbGetQuery(connection, "SELECT COUNT(dest), dest FROM flights GROUP BY dest
           ORDER BY count(dest) DESC LIMIT 5")
```

```
  count(dest) dest
1       17283  ORD
2       17215  ATL
3       16174  LAX
4       15508  BOS
5       14082  MCO
```

Section 1.e: Create a table showing:

- carrier
- average arrival delay (arr_delay)

Sort the results from largest to smallest delay. You must close the connection when you are done. Use `dbDisconnect()`

```
# Group by carriers and calculate the mean arrival delay. Then order
# by the means in descending order.
dbGetQuery(connection, "SELECT carrier, AVG(arr_delay) FROM flights
           GROUP BY carrier ORDER BY AVG(arr_delay) DESC")
```

```
   carrier avg(arr_delay)
1       F9     21.9207048
2       FL     20.1159055
3       EV     15.7964311
4       YV     15.5569853
5       OO     11.9310345
6       MQ     10.7747334
7       WN      9.6491199
8       B6      9.4579733
9       9E      7.3796692
10      UA      3.5580111
11      US      2.1295951
12      VX      1.7644644
13      DL      1.6443409
14      AA      0.3642909
15      HA     -6.9152047
16      AS     -9.9308886
```

```
# Close the database connection
DBI::dbDisconnect(connection)
```

Part II

Section 2.a

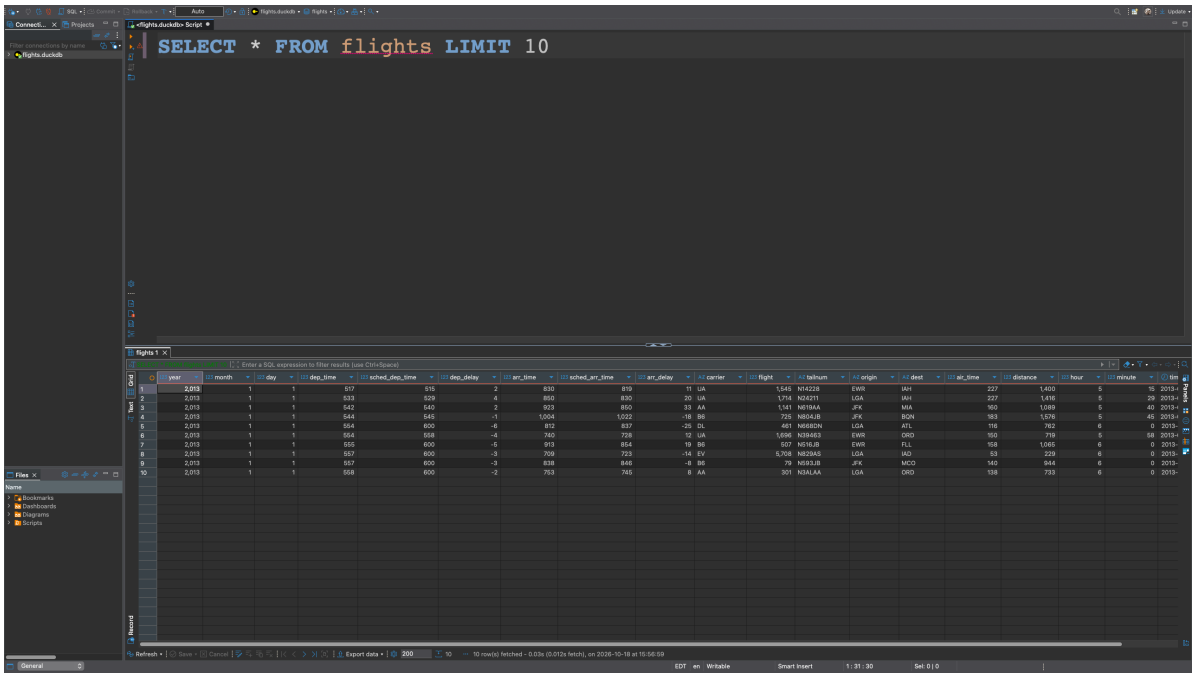


Figure 1: List the first 10 rows of the flights table.

Section 2.b

Section 2.c

Section 2.d

Section 2.e